View CIFS active sessions

ONTAP 9.10.1 REST API Documentation

NetApp
February 13, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi-9101/ontap/protocols_cifs_sessions_endpoint_overview.html on February 13, 2024. Always check docs.netapp.com for the latest.

Table of Contents

View CIFS active sessions . . . 1
Protocols CIFS sessions endpoint overview . . . 1
Retrieve CIFS session information for all SVMs . . . 6
Delete SMB session information for an SVM node . . . 17
Retrieve SMB session information for an SMB connection in an SVM node . . . 19

View CIFS active sessions

Protocols CIFS sessions endpoint overview

Overview

ONTAP CIFS sessions `show` functionality is used to provide a list of currently established CIFS sessions with SMB Clients.

Information on the CIFS session

- List all the SMB sessions for SVM and the clients along with volume information on which the clients are using.

Example

Retrieves established sessions information

To retrieve the list of CIFS sessions, use the following API. Note that *return_records=true*.

```
# The API:
GET /protocols/cifs/sessions

# The call:
curl -X GET "https://<cluster-mgmt-
ip>/api/protocols/cifs/sessions?return_timeout=15&return_records=true" -H
"accept: application/json"

# The response:
{
"records": [
  {
    "node": {
      "uuid": "85d46998-4e5d-11ea-afb1-0050568ec4e4",
      "name": "bkalyan-vsim1"
    },
    "svm": {
      "uuid": "fc824aa8-4e60-11ea-afb1-0050568ec4e4",
      "name": "vs1"
    },
    "identifier": 625718873227788300,
    "connection_id": 91842,
    "lif_address": "10.140.70.197",
    "address": "10.74.7.182",
    "auth_mechanism": "ntlmv2",
```

```
    "windows_user": "NBCIFSQA2\\administrator",
    "unix_user": "root",
    "shares": 1,
    "files": 2,
    "other": 0,
    "connected_time": "PT16H54M47S",
    "idle_time": "PT3S",
    "protocol_version": "smb3_1",
    "continuously_available": "unavailable",
    "is_session_signed": false,
    "smb_encryption_status": "unencrypted",
    "connection_count": 1,
    "is_large_mtu_enabled": true,
    "vol_names": [
      "vol12",
      "origin",
      "fg",
      "vol1"
    ],
    "vol_uuids": [
      "954d697f-4e62-11ea-afb1-0050568ec4e4",
      "0f909e06-4e6e-11ea-afb1-0050568ec4e4",
      "08c27b7d-4e61-11ea-afb1-0050568ec4e4",
      "02f0a240-4e61-11ea-afb1-0050568ec4e4"
    ]
  },
  {
    "node": {
      "uuid": "85d46998-4e5d-11ea-afb1-0050568ec4e4",
      "name": "bkalyan-vsim1"
    },
    "svm": {
      "uuid": "fc824aa8-4e60-11ea-afb1-0050568ec4e4",
       "name": "vs1"
    },
    "identifier": 625718873227788500,
    "connection_id": 92080,
    "lif_address": "10.140.70.197",
    "address": "10.140.133.97",
    "auth_mechanism": "ntlmv2",
    "windows_user": "NBCIFSQA2\\administrator",
    "unix_user": "root",
    "shares": 1,
    "files": 1,
    "other": 0,
    "connected_time": "PT1M19S",
```

```
    "idle_time": "PT1M1S",
    "protocol_version": "smb3",
    "continuously_available": "unavailable",
    "is_session_signed": false,
    "smb_encryption_status": "unencrypted",
    "connection_count": 1,
    "is_large_mtu_enabled": true,
    "vol_names": [
      "origin",
      "fg",
      "vol1"
    ],
    "vol_uuids": [
      "0f909e06-4e6e-11ea-afb1-0050568ec4e4",
      "08c27b7d-4e61-11ea-afb1-0050568ec4e4",
      "02f0a240-4e61-11ea-afb1-0050568ec4e4"
    ]
  }
],
"num_records": 2
}
```

Retrieving CIFS server configuration details for a specific SVM

```
# The API:
GET
/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_i
d}

# The call:
curl -X GET "https://<cluster-mgmt-
ip>/api/protocols/cifs/sessions/85d46998-4e5d-11ea-afb1-
0050568ec4e4/fc824aa8-4e60-11ea-afb1-
0050568ec4e4/625718873227788300/91842?fields=*" -H "accept:
application/json" -H "authorization: Basic YWRtaW46bmV0YXBwMSE="

# The response:
{
"node": {
  "uuid": "85d46998-4e5d-11ea-afb1-0050568ec4e4",
  "name": "bkalyan-vsim1"
},
```

```
"svm": {
  "uuid": "fc824aa8-4e60-11ea-afb1-0050568ec4e4",
  "name": "vs1"
},
"identifier": 625718873227788300,
"connection_id": 91842,
"lif_address": "10.140.70.197",
"address": "10.74.7.182",
"auth_mechanism": "ntlmv2",
"windows_user": "NBCIFSQA2\\administrator",
"unix_user": "root",
"shares": 1,
"files": 2,
"other": 0,
"connected_time": "PT16H54M47S",
"idle_time": "PT3S",
"protocol_version": "smb3_1",
"continuously_available": "unavailable",
"is_session_signed": false,
"smb_encryption_status": "unencrypted",
"connection_count": 1,
"is_large_mtu_enabled": true,
"vol_names": [
  "vol12",
  "origin",
  "fg",
  "vol1"
],
"vol_uuids": [
  "954d697f-4e62-11ea-afb1-0050568ec4e4",
  "0f909e06-4e6e-11ea-afb1-0050568ec4e4",
  "08c27b7d-4e61-11ea-afb1-0050568ec4e4",
  "02f0a240-4e61-11ea-afb1-0050568ec4e4"
]
}
```


Removing all existing CIFS sessions for a specific node on a specific SVM

To delete all the existing CIFS session, pass the identifier and connection ID as zero (0) in the following API. This will delete all of the CIFS sessions for the given SVM on the node.

```
# The API:
DELETE
/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_i
d}

# The call:
curl -X DELETE "https://<cluster-mgmt-
ip>/api/protocols/cifs/sessions/85d46998-4e5d-11ea-afb1-
0050568ec4e4/fc824aa8-4e60-11ea-afb1-0050568ec4e4/0/0" -H "accept:
application/json" -H "authorization: Basic YWRtaW46bmV0YXBwMSE="
```

Removing all CIFS sessions for a specific connection on a specific node on a specific SVM

To delete a CIFS session, use the following API. This will delete the CIFS sessions for a given SVM on the node.

```
# The API:
DELETE
/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_i
d}

# The call:
curl -X DELETE "https://<cluster-mgmt-
ip>/api/protocols/cifs/sessions/85d46998-4e5d-11ea-afb1-
0050568ec4e4/fc824aa8-4e60-11ea-afb1-0050568ec4e4/0/91842" -H "accept:
application/json" -H "authorization: Basic YWRtaW46bmV0YXBwMSE="
```

Removing a specific CIFS session for a specific Node on a specific SVM

To delete a specific CIFS session, use the following API. This will delete the specific CIFS session for the given SVM on the node.

```
# The API:
DELETE
/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_i
d}

# The call:
curl -X DELETE "https://<cluster-mgmt-
ip>/api/protocols/cifs/sessions/85d46998-4e5d-11ea-afb1-
0050568ec4e4/fc824aa8-4e60-11ea-afb1-
0050568ec4e4/625718873227788300/91842" -H "accept: application/json" -H
"authorization: Basic YWRtaW46bmV0YXBwMSE="
```

Retrieve CIFS session information for all SVMs

GET `/protocols/cifs/sessions`

Introduced In: 9.8

Retrieves the CIFS sessions information for all SVMs.

Related ONTAP commands

- `vserver cifs session show -active-volumes`

Learn more

- DOC /protocols/cifs/sessions

Parameters

Name	Type	In	Required	Description
idle_duration	string	query	False	Filter by idle_duration
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name
mapped_unix_user	string	query	False	Filter by mapped_unix_user
open_other	integer	query	False	Filter by open_other

Name	Type	In	Required	Description
smb_encryption	string	query	False	Filter by smb_encryption
connection_id	integer	query	False	Filter by connection_id
open_shares	integer	query	False	Filter by open_shares
open_files	integer	query	False	Filter by open_files
server_ip	string	query	False	Filter by server_ip
identifier	integer	query	False	Filter by identifier
smb_signing	boolean	query	False	Filter by smb_signing
authentication	string	query	False	Filter by authentication
protocol	string	query	False	Filter by protocol
continuous_availability	string	query	False	Filter by continuous_availability
client_ip	string	query	False	Filter by client_ip
volumes.uuid	string	query	False	Filter by volumes.uuid
volumes.name	string	query	False	Filter by volumes.name
connection_count	integer	query	False	Filter by connection_count
user	string	query	False	Filter by user
connected_duration	string	query	False	Filter by connected_duration
node.uuid	string	query	False	Filter by node.uuid

Name	Type	In	Required	Description
node.name	string	query	False	Filter by node.name
large_mtu	boolean	query	False	Filter by large_mtu
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[cifs_session]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "authentication": "ntlmv2",
    "client_ip": "10.74.7.182",
    "connected_duration": "P4DT84H30M5S",
    "connection_count": 0,
    "connection_id": 22802,
    "continuous_availability": "unavailable",
    "identifier": 4622663542519103507,
    "idle_duration": "P4DT84H30M5S",
    "large_mtu": 1,
    "mapped_unix_user": "root",
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "open_files": 0,
    "open_other": 0,
    "open_shares": 0,
    "protocol": "smb3_1",
    "server_ip": "10.140.78.248",
    "smb_encryption": "unencrypted",
    "smb_signing": "",
    "svm": {
      "_links": {
        "self": {
```

```
            "href": "/api/resourcelink"
          }
        },
        "name": "svm1",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "user": "NBCIFSQA2\\administrator",
      "volumes": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "volume1",
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      }
    }
  }
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

volumes

Name	Type	Description
_links	_links	
name	string	The name of the volume.

Name	Type	Description
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • Introduced in: 9.8

cifs_session

Name	Type	Description
_links	_links	
authentication	string	SMB authentication over which the client accesses the share. The following values are supported: • none - No authentication • ntlmv1 - Ntlm version 1 mechanism • ntlmv2 - Ntlm version 2 mechanism • kerberos - Kerberos authentication • anonymous - Anonymous mechanism
client_ip	string	Specifies IP address of the client.
connected_duration	string	Specifies an ISO-8601 format of date and time used to retrieve the connected time duration in hours, minutes, and seconds format.
connection_count	integer	A counter used to track requests that are sent to the volumes to the node.
connection_id	integer	A unique 32-bit unsigned number used to represent each SMB session's connection ID.

Name	Type	Description
continuous_availability	string	The level of continuous availabilty protection provided to the SMB sessions and/or files. • unavailable - Open file is not continuously available. For sessions, it contains one or more open files but none of them are continuously available. • available - open file is continuously available. For sessions, it contains one or more open files and all of them are continuously available. • partial - Sessions only. Contains at least one continuously available open file with other files open but not continuously available.
identifier	integer	A unique 64-bit unsigned number used to represent each SMB session's identifier.
idle_duration	string	Specifies an ISO-8601 format of date and time used to retrieve the idle time duration in hours, minutes, and seconds format.
large_mtu	boolean	Specifies whether the large MTU is enabled or not for an SMB session.
mapped_unix_user	string	Indicated that a mapped UNIX user has logged in.
node	node	
open_files	integer	Number of files the SMB session has opened.
open_other	integer	Number of other files the SMB session has opened.
open_shares	integer	Number of shares the SMB session has opened.

Name	Type	Description
protocol	string	The SMB protocol version over which the client accesses the volumes. The following values are supported: • smb1 - SMB version 1 • smb2 - SMB version 2 • smb2_1 - SMB version 2 minor version 1 • smb3 - SMB version 3 • smb3_1 - SMB version 3 minor version 1
server_ip	string	Specifies the IP address of the SVM.
smb_encryption	string	Indicates an SMB encryption state. The following values are supported: • unencrypted - SMB session is not encrypted • encrypted - SMB session is fully encrypted. SVM level encryption is enabled and encryption occurs for the entire session. • partially_encrypted - SMB session is partially encrypted. Share level encryption is enabled and encryption is initiated when the tree-connect occurs.
smb_signing	boolean	Specifies whether or not SMB signing is enabled.
svm	svm	
user	string	Indicates that a Windows user has logged in.
volumes	array[volumes]	A group of volumes, the client is accessing.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Delete SMB session information for an SVM node

DELETE `/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_id}`

Introduced In: 9.9

Deletes SMB session information on a node for an SVM.

- To delete the specific SMB session information, pass the relavant SMB session's identifier and connection Id.
- To delete all the SMB session information on specific node and SVM, pass the both SMB session's identifier and connection Id as zero(0)
- To delete all the SMB session information on specific connection, pass the specific SMB session's Identifier value as zero(0).
- To delete all the SMB session information on specific Identifier alone is not allowed.

Learn more

- DOC /protocols/cifs/sessions

Parameters

Name	Type	In	Required	Description
node.uuid	string	path	True	Node UUID.

Name	Type	In	Required	Description
identifier	integer	path	True	Unique identifier for the SMB session.
connection_id	integer	path	True	Unique identifier for the SMB connection.
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
655503	The SMB session delete does not allow a connection ID of zero (0).

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve SMB session information for an SMB connection in an SVM node

GET `/protocols/cifs/sessions/{node.uuid}/{svm.uuid}/{identifier}/{connection_id}`

Introduced In: 9.8

Retrieves specific SMB session information for a specific SMB connection in a node on an SVM.

Learn more

- DOC /protocols/cifs/sessions

Parameters

Name	Type	In	Required	Description
node.uuid	string	path	True	Node UUID.
identifier	integer	path	True	Unique identifier for the SMB session.
connection_id	integer	path	True	Unique identifier for the SMB connection.
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	

Name	Type	Description
authentication	string	SMB authentication over which the client accesses the share. The following values are supported: • none - No authentication • ntlmv1 - Ntlm version 1 mechanism • ntlmv2 - Ntlm version 2 mechanism • kerberos - Kerberos authentication • anonymous - Anonymous mechanism
client_ip	string	Specifies IP address of the client.
connected_duration	string	Specifies an ISO-8601 format of date and time used to retrieve the connected time duration in hours, minutes, and seconds format.
connection_count	integer	A counter used to track requests that are sent to the volumes to the node.
connection_id	integer	A unique 32-bit unsigned number used to represent each SMB session's connection ID.

Name	Type	Description
continuous_availability	string	The level of continuous availabilty protection provided to the SMB sessions and/or files. • unavailable - Open file is not continuously available. For sessions, it contains one or more open files but none of them are continuously available. • available - open file is continuously available. For sessions, it contains one or more open files and all of them are continuously available. • partial - Sessions only. Contains at least one continuously available open file with other files open but not continuously available.
identifier	integer	A unique 64-bit unsigned number used to represent each SMB session's identifier.
idle_duration	string	Specifies an ISO-8601 format of date and time used to retrieve the idle time duration in hours, minutes, and seconds format.
large_mtu	boolean	Specifies whether the large MTU is enabled or not for an SMB session.
mapped_unix_user	string	Indicated that a mapped UNIX user has logged in.
node	node	
open_files	integer	Number of files the SMB session has opened.
open_other	integer	Number of other files the SMB session has opened.
open_shares	integer	Number of shares the SMB session has opened.

Name	Type	Description
protocol	string	The SMB protocol version over which the client accesses the volumes. The following values are supported: • smb1 - SMB version 1 • smb2 - SMB version 2 • smb2_1 - SMB version 2 minor version 1 • smb3 - SMB version 3 • smb3_1 - SMB version 3 minor version 1
server_ip	string	Specifies the IP address of the SVM.
smb_encryption	string	Indicates an SMB encryption state. The following values are supported: • unencrypted - SMB session is not encrypted • encrypted - SMB session is fully encrypted. SVM level encryption is enabled and encryption occurs for the entire session. • partially_encrypted - SMB session is partially encrypted. Share level encryption is enabled and encryption is initiated when the tree-connect occurs.
smb_signing	boolean	Specifies whether or not SMB signing is enabled.
svm	svm	
user	string	Indicates that a Windows user has logged in.
volumes	array[volumes]	A group of volumes, the client is accessing.

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": "ntlmv2",
  "client_ip": "10.74.7.182",
  "connected_duration": "P4DT84H30M5S",
  "connection_count": 0,
  "connection_id": 22802,
  "continuous_availability": "unavailable",
  "identifier": 4622663542519103507,
  "idle_duration": "P4DT84H30M5S",
  "large_mtu": 1,
  "mapped_unix_user": "root",
  "node": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "node1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "open_files": 0,
  "open_other": 0,
  "open_shares": 0,
  "protocol": "smb3_1",
  "server_ip": "10.140.78.248",
  "smb_encryption": "unencrypted",
  "smb_signing": "",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "user": "NBCIFSQA2\\administrator",
  "volumes": {
    "_links": {
```

```
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "volume1",
    "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

volumes

Name	Type	Description
_links	_links	
name	string	The name of the volume.
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • Introduced in: 9.8

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.